Marcelo d'Amorim

Associate Professor, Computer Science Department
North Carolina State University

Address
890 Oval Drive
3320 Engineering Building II
Raleigh, NC 27606

Email ..... mdamori@ncsu.edu
Personal Page ..... https://damorim.github.io
Work Phone ..... +1 (919) 515-3644

Education

University of Illinois at Urbana-Champaign ..... Champaign, IL, US
2007 Ph.D. in Computer Science
Dissertation title: "Efficient Explicit-State Model Checking for Programs with Dynamically Allocated Data"
Advisor: Darko Marinov

Federal University of Pernambuco ..... Recife, PE, Brazil
2001 M.S. in Computer Science
1997 B.S. in Computer Science

Research Interests

My research interests are in the areas of **Software Engineering and Programming Languages**, with a focus on improving software reliability through program analysis and systematic testing.

Experience

22– **North Carolina State University** ..... Raleigh, NC, US
Associate Professor. Advising: 6 PhDs; Graduated: 1 PhD

09–22 **Federal University of Pernambuco (UFPE)** ..... Recife, PE, Brazil
Associate Professor. Graduated: 2 PhD + 12 MS

15–16 **Georgia Institute of Technology** ..... Atlanta, GA, US
Visiting Scholar at the Arktos group led by Alessandro Orso

07–08 **Federal University of Pernambuco** ..... Recife, PE, Brazil
Postdoctoral researcher at the SPG group led by Paulo Borba

Awards

2026 IEEE Computer Society TCSE Best Student Paper Award (ICST), for [C2]
2025 ACM Foundations of Software Engineering (FSE) Distinguished Reviewer Award
2020 Best-paper nominee at IEEE ICST [C19]; invited for journal extension [J12]
2016 IEEE/ACM Automated Software Engineering (ASE) Distinguished Reviewer Award
2014 Finalist (with 5 others), Microsoft Research Faculty Fellowship Latin America
2013 Microsoft Software Engineering Innovation Foundation (SEIF) Award
2010 Best-paper nominee at CBSoft/SBLP; invited for journal extension [J18]
2009 Best-paper nominee at NASA Formal Methods (NFM) [C39]; invited for journal extension [J20]
2007 Best-paper nominee at ACM ISSTA [C41]; invited for journal extension [J21]
2006 Best-paper nominee at IEEE/ACM ASE [C43]; invited for journal extension

Funding

Table 1 shows the awarded grants since I joined NC State in November 2022. Each row of the table corresponds to one grant. The columns show, from left to right, the id (titles listed below the table), the sponsor, the budget period, my role, and the total amount of the grant. The row in gray color shows a grant transferred to me when a colleague left NC State. Together, these grants total **$1,433,498**.

Table 1: Awarded grants sorted in reverse chronological order of end date.

Id	Sponsor	Budget Period	Role	Total
1	NSF	07/01/2024 - 06/30/2027	PI	$572,778
2	NSF	09/01/2023 - 02/28/2026	PI	$100,000
3	L3Harris	08/19/2024 - 12/31/2025	Co-PI	$316,000
4	NSF	10/01/2020 - 09/30/2025	PI	$444,720

1. SHF: Small: E2R2: A Comprehensive Approach to Improve Simulation-based Testing of Autonomous Driving Systems
2. Collaborative Research: FMitF: Track II: Cross-Language Support for Runtime Verification
3. Software Defect Discovery
4. Collaborative Research: SaTC: TTP: Small: eSLIC: Enhanced Security Static Analysis for Detecting Insecure Configuration Scripts

Publications

3,274 citations, h-index 35, i10-index 58, as of July 2026 (Google Scholar).

Refereed Conference (Full) Papers

[C1] F. Molina, **M. M. A. Naziri**, **F. Qin**, A. Gorla, and M. d'Amorim, "Improving Deep Learning Library Testing with Machine Learning," in *Proceedings of the ACM/IEEE International Conference on Automation of Software Test (AST)*, 2026.

[C2] **F. Qin**, **M. M. A. Naziri**, H. Ai, S. Dutta, and M. d'Amorim, "How Effective Is Coverage-Guided Fuzzing to Test Deep Learning Library APIs?" In *Proceedings of the IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2026.

[C3] P. Jiang, K. Guan, M. Khosravi, M. Ismail, M. d'Amorim, and O. Legunsen, "Fine-Grained Analyses for Evolution-Aware Runtime Verification," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026.

[C4] **M. M. A. Naziri**, S. C. Lambertenghi, A. Stocco, and M. d'Amorim, "Misbehavior Forecasting for Focused Autonomous Driving Systems Testing," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026.

[C5] **M. M. A. Naziri**, S. Kim, **F. Qin**, S. Dutta, and M. d'Amorim, "Testing Deep Learning Libraries via Neurosymbolic Constraint Learning," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026.

[C6] D. Silva, **M. Farahat**, and M. d'Amorim, "An Empirical Analysis of Cross-OS Portability Issues in Python Projects," in *Proceedings of the International Conference on Mining Software Repositories (MSR)*, 2026.

[C7] **L. Liu**, **F. Qin**, O. Legunsen, and M. d'Amorim, "Learning Compiler Fuzzing Mutators from Historical Bugs," in *Proceedings of the International Conference on Mining Software Repositories (MSR)*, 2026.

[C8] K. Guan, M. d'Amorim, et al., "Faster Explicit-Trace Monitoring-Oriented Programming for Runtime Verification of Software Tests," in *Proceedings of the ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2025.

[C9] **U. Kulsum**, H. Zhu, B. Xu, and M. d'Amorim, "A Case Study of LLM for Automated Vulnerability Repair: Assessing Impact of Reasoning and Patch Validation Feedback," in *Proceedings of the 1st ACM International Conference on AI-Powered Software, AIware 2024, Porto de Galinhas, Brazil, July 15-16, 2024*, 2024.

[C10] **I. Hayet**, **A. Scott**, and M. d'Amorim, "Feedback-Directed Partial Execution," in *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2024, pp. 781–793.

[C11] R. Lima, K. Costa, J. Souza, L. Teixeira, B. Fonseca, M. d'Amorim, M. Ribeiro, and B. Miranda, "Do You See Any Problem? On the Developers Perceptions in Test Smells Detection," in *Proceedings of the XXII Brazilian Symposium on Software Quality, SBQS 2023, Brasilia, Brazil, November 7-10, 2023*, 2023, pp. 21–30.

[C12] A. Stocco, P. J. Nunes, M. d'Amorim, and P. Tonella, "ThirdEye: Attention Maps for Safe Autonomous Driving Systems," in *Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2022, 102:1–102:12.

[C13] S. Reis, R. Abreu, M. d'Amorim, and D. Fortunato, "Leveraging Practitioners' Feedback to Improve a Security Linter," in *Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2022, 66:1–66:12.

[C14] F. Molina, M. d'Amorim, and N. Aguirre, "Fuzzing Class Specifications," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2022, pp. 1008–1020.

[C15] S. Mondal, D. Silva, and M. d'Amorim, "Soundy Automated Parallelization of Test Execution," in *IEEE International Conference on Software Maintenance and Evolution, ICSME 2021, Luxembourg, September 27 - October 1, 2021*, 2021, pp. 309–319.

[C16] L. Teixeira, B. Miranda, H. Rebêlo, and M. d'Amorim, "Demystifying the Challenges of Formally Specifying API Properties for Runtime Verification," in *Proceedings of the IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2021, pp. 82–93.

[C17] J. Henkel, D. Silva, L. Teixeira, M. d'Amorim, and T. W. Reps, "Shipwright: A Human-in-the-Loop System for Dockerfile Repair," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2021, pp. 1148–1160.

[C18] D. Silva, L. Teixeira, and M. d'Amorim, "Shake It! Detecting Flaky Tests Caused by Concurrency with Shaker," in *IEEE International Conference on Software Maintenance and Evolution, ICSME 2020, Adelaide, Australia, September 28 - October 2, 2020*, 2020, pp. 301–311.

Marked text indicates work done at NC State.
Authors in bold are/were NC State students.

[C19] B. Miranda, I. Lima, O. Legunsen, and M. d'Amorim, "Prioritizing Runtime Verification Violations," in *Proceedings of the IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2020, pp. 297–308.

[C20] M. A. Guimarães, L. Fernandes, M. Ribeiro, M. d'Amorim, and R. Gheyi, "Optimizing Mutation Testing by Discovering Dynamic Mutant Subsumption Relations," in *Proceedings of the IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2020, pp. 198–208.

[C21] G. Pinto, B. Miranda, S. Dissanayake, M. d'Amorim, C. Treude, and A. Bertolino, "What is the Vocabulary of Flaky Tests?" In *Proceedings of the International Conference on Mining Software Repositories (MSR)*, 2020, pp. 492–502.

[C22] S. Reis, R. Abreu, and M. d'Amorim, "Demystifying the Combination of Dynamic Slicing and Spectrum-based Fault Localization," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2019.

[C23] X. Li, M. d'Amorim, and A. Orso, "Intent-Preserving Test Repair," in *Proceedings of the IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2019, pp. 217–227.

[C24] M. Fazzini, M. Prammer, M. d'Amorim, and A. Orso, "Automatically Translating Bug Reports into Test Cases for Mobile Apps," in *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2018, pp. 141–152.

[C25] X. Li, S. Zhu, M. d'Amorim, and A. Orso, "Enlightened Debugging," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2018, pp. 82–92.

[C26] A. Perez, R. Abreu, and M. D'Amorim, "Prevalence of Single-Fault Fixes and Its Impact on Fault Localization," in *Proceedings of the IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2017, pp. 12–22.

[C27] J. Candido, L. Melo, and M. d'Amorim, "Test Suite Parallelization in Open-Source Projects: A Study on Its Usage and Impact," in *Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2017, pp. 838–848.

[C28] S. Souto, M. d'Amorim, and R. Gheyi, "Balancing Soundness and Efficiency for Practical Testing of Configurable Systems," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2017, pp. 632–642.

[C29] X. Li, M. d'Amorim, and A. Orso, "Iterative User-Driven Fault Localization," in *Hardware and Software: Verification and Testing - 12th International Haifa Verification Conference, HVC 2016, Haifa, Israel, November 14-17, 2016, Proceedings*, 2016, pp. 82–98.

[C30] M. Borges, A. Filieri, M. d'Amorim, and C. S. Păsăreanu, "Iterative Distribution-aware Sampling for Probabilistic Symbolic Execution," in *Proceedings of the 10th Joint Meeting on Foundations of Software Engineering*, 2015, pp. 866–877.

[C31] P. Barros, R. Just, S. Millstein, P. Vines, W. Dietl, M. d'Amorim, and M. D. Ernst, "Static Analysis of Implicit Control Flow: Resolving Java Reflection and Android Intents," in *Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2015, pp. 669–679.

[C32] T. Liu, M. Araújo, M. d'Amorim, and M. Taghdiri, "A Comparative Study of Incremental Constraint Solving Approaches in Symbolic Execution," in *Proceedings of the 10th International Haifa Verification Conference*, 2014, pp. 284–299.

[C33] M. Borges, A. Filieri, M. d'Amorim, C. S. Păsăreanu, and W. Visser, "Compositional Solution Space Quantification for Probabilistic Software Analysis," in *Proceedings of the 35th ACM SIGPLAN Conference on Programming Language Design and Implementation*, 2014, pp. 123–132.

[C34] C. H. P. Kim, D. Marinov, S. Khurshid, D. Batory, S. Souto, P. Barros, and M. d'Amorim, "SPLat: Lightweight Dynamic Analysis for Reducing Combinatorics in Testing Configurable Systems," in *Proceedings of the 9th Joint Meeting on Foundations of Software Engineering*, 2013, pp. 257–267.

[C35] J. Campos, R. Abreu, G. Fraser, and M. d'Amorim, "Entropy-based test generation for improved fault localization," in *Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2013, pp. 257–267.

[C36] M. Borges, M. d'Amorim, S. Anand, D. H. Bushnell, and C. S. Pasareanu, "Symbolic Execution with Interval Solving and Meta-heuristic Search," in *Proceedings of the IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2012, pp. 111–120.

[C37] M. Souza, M. Borges, M. d'Amorim, and C. S. Pasareanu, "CORAL: Solving Complex Constraints for Symbolic PathFinder," in *NASA Formal Methods - Third International Symposium, NFM 2011, Pasadena, CA, USA, April 18-20, 2011. Proceedings*, 2011, pp. 359–374.

[C38] A. Rimsa, M. d'Amorim, and F. M. Q. Pereira, "Tainted Flow Analysis on e-SSA-form Programs," in *Proceedings of the 20th International Conference on Compiler Construction: Part of the Joint European Conferences on Theory and Practice of Software*, 2011, pp. 124–143.

[C39] M. Takaki, D. Cavalcanti, R. Gheyi, J. Iyoda, M. d'Amorim, and R. B. C. Prudêncio, "A Comparative Study of Randomized Constraint Solvers for Random-Symbolic Testing," in *First NASA Formal Methods Symposium - NFM 2009, Moffett Field, California, USA, April 6-8, 2009.*, 2009, pp. 56–65.

[C40] C. Bertolini, G. Peres, M. d'Amorim, and A. Mota, "An Empirical Evaluation of Automated Black Box Testing Techniques for Crashing GUIs," in *Proceedings of the IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2009, pp. 21–30.

[C41] M. d'Amorim, S. Lauterburg, and D. Marinov, "Delta Execution for Efficient State-space Exploration of Object-oriented Programs," in *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2007, pp. 50–60.

[C42] M. d'Amorim, A. Sobeih, and D. Marinov, "Optimized Execution of Deterministic Blocks in Java Pathfinder," in *Proceedings of the 8th International Conference on Formal Methods and Software Engineering*, 2006, pp. 549–567.

[C43] M. d'Amorim, C. Pacheco, T. Xie, D. Marinov, and M. D. Ernst, "An Empirical Comparison of Automated Generation and Classification Techniques for Object-Oriented Unit Testing," in *Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2006, pp. 59–68.

[C44] M. d'Amorim and G. Roşu, "Efficient Monitoring of ω-Languages," in *Proceedings of 17th International Conference on Computer-aided Verification*, vol. 3576, 2005, pp. 364–378.

[C45] F. Chen, M. d'Amorim, and G. Rosu, "A Formal Monitoring-Based Framework for Software Development and Analysis," in *Formal Methods and Software Engineering, 6th International Conference on Formal Engineering Methods, ICFEM 2004, Seattle, WA, USA, November 8-12, 2004, Proceedings*, vol. 3308, 2004, pp. 357–372.

Journal Papers

[J1] X. Chen, M. Biagiola, V. Riccio, M. d'Amorim, and A. Stocco, "XMutant: XAI-based Fuzzing for Deep Learning Systems," *Empirical Software Engineering*, vol. 31, no. 4, p. 102, 2026.

[J2] R. Melo, P. Simões, R. Gheyi, M. d'Amorim, M. Ribeiro, G. Soares, E. Almeida, and E. Soares, "Agentic LMs: Hunting Down Test Smells," *IEEE Software*, vol. 43, no. 1, pp. 32–40, 2026.

[J3] Y. Zhang, **U. Paul**, M. d'Amorim, and A. Rahman, "Configuration Defects in Kubernetes," *IEEE Transactions on Software Engineering*, vol. 52, no. 3, pp. 1125–1144, 2026.

[J4] **I. Hayet**, **A. Scott**, and M. d'Amorim, "ChatAssert: LLM-based Test Oracle Generation with External Tools Assistance," *IEEE Transactions on Software Engineering*, no. 01, pp. 1–18, 2025.

[J5] F. Molina and A. Gorla, "Test Oracle Automation in the Era of LLMs," *ACM Transactions on Software Engineering and Methodology*, 2025.

[J6] L. Pasquale, A. Sabetta, M. d'Amorim, P. Hegedüs, M. T. Mirakhorli, H. Okhravi, M. Payer, A. Rashid, J. C. S. Santos, J. M. Spring, L. Tan, and K. Tuma, "Challenges to Using Large Language Models in Code Generation and Repair," *IEEE Security & Privacy*, vol. 23, no. 2, pp. 81–88, 2025.

[J7] D. Silva, M. Gruber, S. Gokhale, E. Arteca, A. Turcotte, M. d'Amorim, W. Lam, S. Winter, and J. Bell, " The Effects of Computational Resources on Flaky Tests," *IEEE Transactions on Software Engineering*, no. 01, pp. 1–18, 2024.

[J8] K. Barbosa, R. Ferreira, G. Pinto, M. d'Amorim, and B. Miranda, "Test Flakiness Across Programming Languages," *IEEE Trans. Software Eng.*, vol. 49, no. 4, pp. 2039–2052, 2023.

[J9] B. Reid, M. d'Amorim, M. Wagner, and C. Treude, "NCQ: Code Reuse Support for Node.js Developers," *IEEE Trans. Software Eng.*, vol. 49, no. 5, pp. 3205–3225, 2023.

[J10] A. Torres, P. H. T. Costa, L. H. V. Amaral, J. Pastro, R. Bonifácio, M. d'Amorim, O. Legunsen, E. Bodden, and E. D. Canedo, "Runtime Verification of Crypto APIs: An Empirical Study," *IEEE Trans. Software Eng.*, vol. 49, no. 10, pp. 4510–4525, 2023.

[J11] L. Alcantara, G. Padilha, R. Abreu, and M. d'Amorim, "Syrius: Synthesis of Rules for Intrusion Detectors," *IEEE Trans. Reliab.*, vol. 71, no. 1, pp. 370–381, 2022.

[J12] L. Cabral, B. Miranda, I. Lima, and M. d'Amorim, "RVprio: A Tool for Prioritizing Runtime Verification Violations," *Softw. Test. Verification Reliab.*, vol. 32, no. 5, 2022.

[J13] R. Gheyi, M. Ribeiro, B. Souza, M. A. Guimarães, L. Fernandes, M. d'Amorim, V. Alves, L. Teixeira, and B. Fonseca, "Identifying Method-Level Mutation Subsumption Relations Using Z3," *Inf. Softw. Technol.*, vol. 132, p. 106 496, 2021.

[J14] I. Lima, J. Silva, B. Miranda, G. Pinto, and M. d'Amorim, "Exposing Bugs in JavaScript Engines Through Test Transplantation and Differential Testing," *Softw. Qual. J.*, vol. 29, no. 1, pp. 129–158, 2021.

Marked text indicates work done at NC State.
Authors in bold are/were NC State students.

[J15] L. Melo, I. Wiese, and M. d'Amorim, "Using Docker to Assist Q&A Forum Users," *IEEE Trans. Software Eng.*, vol. 47, no. 11, pp. 2563–2574, 2021.

[J16] I. Lima, J. Cândido, and M. d'Amorim, "Practical Detection of CMS Plugin Conflicts in Large Plugin Sets," *Inf. Softw. Technol.*, vol. 118, 2020.

[J17] S. Souto and M. d'Amorim, "Time-Space Efficient Regression Testing for Configurable Systems," *Journal of Systems and Software*, vol. 137, pp. 733–746, 2018.

[J18] A. Rimsa, M. d'Amorim, F. M. Q. Pereira, and R. S. Bigonha, "Efficient Static Checker for Tainted Variable Attacks," *Science of Computer Programming*, vol. 80, pp. 91–105, 2014.

[J19] A. Sobeih, M. d'Amorim, M. Viswanathan, D. Marinov, and J. C. Hou, "Assertion Checking in J-Sim Simulation Models of Network Protocols," *Simulation*, vol. 86, no. 11, pp. 651–673, 2010.

[J20] M. Takaki, D. Cavalcanti, R. Gheyi, J. Iyoda, M. d'Amorim, and R. B. C. Prudêncio, "Randomized Constraint Solvers: A Comparative Study," *Innovations in Systems and Software Engineering*, vol. 6, no. 3, pp. 243–253, 2010.

[J21] M. d'Amorim, S. Lauterburg, and D. Marinov, "Delta Execution for Efficient State-Space Exploration of Object-Oriented Programs," *IEEE Transactions on Software Engineering*, vol. 34, no. 5, pp. 597–613, 2008.

[J22] F. Chen, M. d'Amorim, and G. Roşu, "Checking and Correcting Behaviors of Java Programs at Runtime with Java-MOP," *Electronic Notes in Theoretical Computer Science*, vol. 144, no. 4, pp. 3–20, 2006.

[J23] M. d'Amorim and K. Havelund, "Event-based Runtime Verification of Java Programs," *SIGSOFT Software Engineering Notes*, vol. 30, no. 4, pp. 1–7, 2005.

Short Papers (Conference and Workshop)

[S1] Z. Shen, M. Yaseen, K. Guan, D. Silva, M. d'Amorim, and O. Legunsen, "PyMOP: A Runtime Verification Tool for Python," in *Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE): Tool Demonstrations*, 2026.

[S2] **S. Elder**, A. Klevans, R. Paramitha, M. d'Amorim, and L. Williams, "Aggregating Security Measures from the Dependency Tree," in *Proceedings of the ACM Workshop on Secure and Trustworthy Software Supply Chain (SCORED)*, 2025.

[S3] **M. M. A. Naziri**, S. Dutta, and M. d'Amorim, "BugsInDLLs: A Database of Reproducible Bugs in Deep Learning Libraries to Enable Systematic Evaluation of Testing Techniques," in *Proceedings of the 34th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA Companion)*, 2025, pp. 1–5.

[S4] **S. Hamer**, M. d'Amorim, and L. A. Williams, "Just Another Copy and Paste? Comparing the Security Vulnerabilities of ChatGPT Generated Code and StackOverflow Answers," in *IEEE Security and Privacy, SP 2024 - Workshops, San Francisco, CA, USA, May 23, 2024*, 2024, pp. 87–94.

[S5] F. Molina, M. d'Amorim, and N. Aguirre, "SpecFuzzer: A Tool for Inferring Class Specifications via Grammar-Based Fuzzing," in *Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2023, pp. 2094–2097.

[S6] B. Reid, M. Wagner, M. d'Amorim, and C. Treude, "Software Engineering User Study Recruitment on Prolific: An Experience Report," in *International Workshop on Recruiting Participants for Empirical Software Engineering (RoPES)*. 2022.

[S7] M. Cordeiro, D. Silva, L. Teixeira, B. Miranda, and M. d'Amorim, "Shaker: a Tool for Detecting More Flaky Tests Faster," in *Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2021, pp. 1281–1285.

[S8] D. S. Costa, C. A. B. Mello, and M. d'Amorim, "A Comparative Study on Methods and Tools for Handwritten Mathematical Expression Recognition," in *DocEng '21: ACM Symposium on Document Engineering 2021, Limerick, Ireland, August 24-27, 2021*, 2021, 26:1–26:4.

[S9] M. d'Amorim, R. Abreu, and C. Mello, "Visual Sketching: From Image Sketches to Code," in *ICSE-NIER 2020: 42nd International Conference on Software Engineering, New Ideas and Emerging Results, Seoul, South Korea, 27 June - 19 July, 2020*, 2020, pp. 101–104.

[S10] D. M. Junior, L. Melo, H. Lu, M. d'Amorim, and A. Prakash, "A Study of Vulnerability Analysis of Popular Smart Devices Through Their Companion Apps," in *IEEE Workshop on the Internet of Safe Things (SafeThings)*, 2019.

[S11] Q.-S. Phan, P. Malacaria, C. S. Păsăreanu, and M. d'Amorim, "Quantifying Information Leaks Using Reliability Analysis," in *Proceedings of the International SPIN Symposium on Model Checking of Software*, 2014, pp. 105–108.

Marked text indicates work done at NC State.
Authors in bold are/were NC State students.

[S12] E. Alves, M. Gligoric, V. Jagannath, and M. d'Amorim, "Fault-Localization Using Dynamic Slicing and Change Impact Analysis," in *Proceedings of the IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2011, pp. 520–523.

[S13] T. Gvero, M. Gligoric, S. Lauterburg, M. d'Amorim, D. Marinov, and S. Khurshid, "State Extensions for Java PathFinder," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2008, pp. 863–866.

[S14] Y. Zhou, D. Marinov, W. Sanders, C. Zilles, M. d'Amorim, S. Lauterburg, R. M. Lefever, and J. Tucek, "Delta Execution for Software Reliability," in *Proceedings of the 3rd Workshop on on Hot Topics in System Dependability*, 2007, 16–es.

[S15] F. Chen, M. d'Amorim, and G. Rosu, "Checking and Correcting Behaviors of Java Programs at Runtime with Java-MOP," in *Proceedings of the Fifth Workshop on Runtime Verification, RV@CAV 2005, Edinburgh, UK, July 12, 2005*, vol. 144, 2005, pp. 3–20.

[S16] M. d'Amorim and K. Havelund, "Event-based Runtime Verification of Java Programs," in *Proceedings of the Third International Workshop on Dynamic Analysis*, 2005, pp. 1–7.

[S17] M. d'Amorim, C. Nogueira, G. Santos, A. Souza, and P. Borba, "Integrating Code Generation and Refactoring," in *Workshop on Generative Programming, 16th European Conference on Object-Oriented Programming (ECOOP 2002)*. 2002, pp. 1–11.

[S18] M. d'Amorim and C. A. G. Ferraz, "A Design for JTrader, an Internet Trading Service," in *Innovative Internet Computing Systems, International Workshop IICS 2001, Ilmenau, Germany, June 21-22, 2001, Proceedings*, vol. 2060, 2001, pp. 159–166.

Edited Proceedings

[P1] *Companion Proceedings of the 32nd ACM International Conference on the Foundations of Software Engineering (FSE 2024)*, 2024.

Student Advising

Most of my advising in Brazil was at the Master's level: I graduated 12 Master's students and 2 PhD students at UFPE. I have been building a PhD pipeline since joining NC State, and the six students I currently advise were all admitted from 2022 onward. All three of my PhD graduates hold faculty positions, in the United States and in Brazil.

I also work to draw the students in my classes into research, and I take part in UIUC++ SRSE (Summer Research in Software Engineering, `https://uiuc-srse.github.io`), through which I mentor undergraduates seeking a first experience in research. Many of these efforts have resulted in publications, including [C2, C3, C16, J2, J8, J11].

Graduated PhD Students

PhD'26 Denini Silva (UFPE)
Detection and Prevention of Resource-Aware Flaky Tests
First job: Rural Federal University of Pernambuco (Brazil)

PhD'24 Sarah Elder (NC State, Co-advised with Laurie Williams)
Title: Supply-Chain Security Analysis with Score Cards
First job: University of North Carolina Wilmington

PhD'15 Sabrina Souto (UFPE)
Addressing High Dimensionality and Lack of Feature Models in Testing of Software Product Lines
First job: State University of Paraíba (Brazil)

Graduated MS Students: 13

Current PhD Students: 6 (`https://damorim.github.io/students.html`)

Marked text indicates work done at NC State.
Authors in bold are/were NC State students.

Software and Industrial Impact

My students and I develop general testing methods and apply them to infrastructure tools that much of the software industry depends on. These include, for example, PyTorch and TensorFlow, which underpin modern AI research and production, and GCC and LLVM, which compile much of the world's software. A defect in any of these platforms propagates into thousands of downstream systems, so fixing it at the source multiplies the impact.

Representative tools appear below, most recent first. Other tools: https://github.com/ncsu-swat.

FlashFuzz [C2] (2026) is an LLM-based test harness generator for deep learning libraries. Given an API and its documentation, it synthesizes a harness that turns the byte strings a coverage-guided fuzzer manipulates into valid API inputs, and repairs broken harnesses from execution feedback. The result is a harness that an off-the-shelf fuzzer drives to expose crashes. FlashFuzz found **54 previously unknown bugs in PyTorch and TensorFlow**; 53 are confirmed and 15 already fixed.

Centaur [C5] (2026) infers input specifications for deep learning library APIs. From a small set of automatically generated seed inputs, it learns first-order-logic constraints on an API's inputs, then solves them with an SMT solver to produce valid and diverse test inputs. Centaur found **26 new bugs in PyTorch and TensorFlow**, 18 of them confirmed.

IssueMut [C7] (2026) mines compiler fuzzing mutators from bug histories. Given a corpus of GCC and LLVM bug reports together with seed programs, it derives mutators that are retrofitted into an existing mutational fuzzer, which in turn generates bug-triggering programs. IssueMut reported **65 bugs in GCC and LLVM**, 60 of them confirmed or fixed.

Foresee [C4] (2026) steers simulation-based testing of autonomous driving systems toward near misses, on the insight that a near miss often precedes a real failure. Given a driving scenario and the system under test, a forecaster computes possible future states of the ego-vehicle to flag such cases, and local fuzzing around them yields failure-inducing scenarios. Foresee exposes **128.7% more failures** than a random baseline and lifts DriveFuzz's detection by up to **93.9%**.

qCORAL [C30, C33] (2014–2015) quantifies solution spaces for probabilistic program analysis. Given a numeric path condition and probability distributions over the program's inputs, it estimates the probability that the condition holds, together with a confidence interval, by decomposing the condition into independent parts and sampling each in a distribution-aware manner. qCORAL **quantifies arbitrarily complex constraints and continuous distributions**, which prior approaches could handle only if linear or discretized, improving on them in both scalability and accuracy.

CORAL [C36, C37] (2011–2012) is a constraint solver for the numeric path conditions that symbolic execution produces on floating-point code. It takes a path condition over floating-point variables, including transcendental functions that defeat conventional decision procedures, and returns a satisfying assignment found by combining interval solving with meta-heuristic search. CORAL **efficiently solves constraints that are expensive for traditional solvers** and **is used as a constraint solver in NASA's Symbolic PathFinder**, broadening the reach of symbolic execution at NASA and in industry.

Teaching

My teaching centers on Software Engineering, Software Testing, and Compilers. I have taught Software Testing at both the undergraduate and graduate levels. My teaching typically pairs a theoretical component with a practical one. The pairing is deliberate: the practical component builds fluency with the tools the discipline uses today, while the theoretical component grounds students in the principles that outlast them (e.g., specification, abstraction, and rigorous reasoning about program behavior). I revise the syllabus every semester, guided both by the questions the field is actively arguing about (e.g., the role of agentic AI in software construction) and by the feedback I receive from my students.

Service

I served as **General Chair of ACM ESEC/FSE 2024**, as Tutorials Co-Chair of ESEC/FSE 2023, and I have been a member of the ESEC/FSE Steering Committee since 2021. Between 2014 and 2026 I served on **41 conference program committees**: 24 at the four leading software engineering venues, namely ICSE (8), ISSTA (8), ASE (5), and ESEC/FSE (3); 7 at other international venues, namely ICST (4), ICSE SEIP (2), and ICSE NIER (1); and 10 at Brazilian venues, namely SBCARS (3), SBES (3), SBLP (2), CBSoft (1), and SAST (1). I served on 13 further program committees for workshops and non-main tracks. Other roles include Program Co-Chair of TAIC-PART 2020, chair of the award committees for the ICST Most Influential Paper (2018) and the ACM SIGSOFT Impact Paper Award (2015), and guest co-editor of the Journal of Software Testing, Verification and Reliability (STVR) since 2018. The complete list is available at https://damorim.github.io/service.html.